OUTLINE PARA SA PAGPAPLANO SA PAMPANAHONANG PAPEL TUNGO SA PAGBUOD NG PROPOSAL PAPER

(BSED MATH 1B)

I. Pagpili at Paglilimita ng Paksa (Discipline-based)

B. BSED MATHEMATICS

Math anxiety, Mathematics Performance

II. Pagbuo ng Rasyunal ng Pag-aaral

- Ano ang problema sa larangan?

  - Mababa ang performance ng mga mag-aaral sa Math dahil sa anxiety.

- Bakit ito mahalaga sa propesyon?

  - Mahalaga ito sa mga guro upang makagawa ng epektibong estratehiya sa pagtuturo na makababawas ng anxiety at makapagpapaangat ng learning outcomes.

- Sino ang makikinabang?

  1. Mag aaral- tataas ang kumpiyansa at performance sa Math
  2. Guro- gabay sa mas epektibong teaching strategies
  3. Komunidad- mas handang kabataan sa academic challenges.
  4. Industriya- mas maraming indibidwal na may kasanayan sa numeracy

- Ano ang puwang sa kaalaman na nais punan?

  - Kulang pa ang lokal na pag-aaral tungkol sa ugnayan ng math anxiety at math performance ng mga estudyante, lalo na sa konteksto ng mga Pilipinong mag-aaral.

III. Pagbuo ng Layunin at Tanong sa Pananaliksik

• Pangkalahatang Layunin - Upang masuri ang ugnayan sa pagitan ng Math Anxiety at ng Math Performance ng mga mag-aaral sa BSEd Math 2nd Year sa asignaturang Matematika.

• Mga tanong

1. Ano ang antas ng Math Anxiety ng mga mag aaral sa ikalawang taon ng kolehiyo batay sa mga sumusunod na salik?
   a) panlipunang salik (social factors)
   b) salik na may kaugnayan sa pagtataya o pagsusulit (assessment factors), at
   c) pisikal at emosyonal na salik (physical and emotional factors)
2. Ano ang antas ng Math Performance ng mga mag aaral sa ikalawang taon ng kolehiyo?
3. Ano ang ugnayan ng Math Anxiety sa Mathematics Performance ng mga BSEd Math second year college students?

IV. Preliminary Review of Related Literature

Math Anxiety

a) Panlipunang salik (social factors)

Zanabazar, A., Deleg, A., & Ravdan, M. (2023). A study of factors causing math anxiety among undergraduate students. International Journal of Innovative Research and Scientific Studies, 6(3), 578–585. https://doi.org/10.53894/ijirss.v6i3.1609

Hossain, M. K., Shimul, M. M. H., Talukder, A. S., & Islam, M. M. (2025). Impact of math anxiety and general anxiety on academic performance among undergraduates in Bangladesh. Acta Psychologica, 260, 105620. https://doi.org/10.1016/j.actpsy.2025.105620

Mizuhara, M. S., Toms, K., & Williams, M. (2024). A bounded confidence model to predict how group work affects student math anxiety. arXiv (Cornell University). https://doi.org/10.48550/arxiv.2407.06351

Ciringione, L., Franchino, E., Reigl, S., D'Onofrio, I., Serbati, A., Poquet, O., Gabriel, F., & Stella, M. (2025). Math anxiety and associative knowledge structure are entwined in psychology students but not in Large Language Models like GPT-3.5 and GPT-4o. arXiv (Cornell University). https://doi.org/10.48550/arxiv.2511.01558

Rada, E., & Lucietto, A. M. (2022). Math anxiety: A literature review on confounding factors. Journal of Research in Science,

Mathematics and Technology Education, 5(2), 117–129. https://doi.org/10.31756/jrsmte.12040

Demedts, F., Reynvoet, B., Sasanguie, D., & Depaepe, F. (2022). Unraveling the role of math anxiety in students' math performance. Frontiers in Psychology, 13, 979113. https://doi.org/10.3389/fpsyg.2022.979113

Barroso, C., Ganley, C. M., McGraw, A. L., Geer, E. A., Hart, S. A., & Daucourt, M. C. (2020). A meta-analysis of the relation between math anxiety and math achievement. Psychological Bulletin, 147(2), 134–168. https://doi.org/10.1037/bul0000307

b) Salik na may kaugnayan sa pagtataya o pagsusulit (assessment factors), at

Rada, E., & Lucietto, A. M. (2022). Math anxiety: A literature review on confounding factors. Journal of Research in Science, Mathematics and Technology Education, 5(2), 117–129. https://doi.org/10.31756/jrsmte.12040

Ramirez, G., Shaw, S. T., & Maloney, E. A. (2018). Math anxiety: Past research, promising interventions, and a new interpretation framework. Educational Psychologist, 53(3), 145–164. https://doi.org/10.1080/00461520.2018.1447384

Supekar, K., Iuculano, T., Chen, L., & Menon, V. (2015). Remediation of childhood math anxiety and associated neural circuits through cognitive tutoring. Journal of Neuroscience, 35(36), 12574–12583. https://doi.org/10.1523/JNEUROSCI.0786-15.2015

Viray-Ariola, C. A. (2025). College students' level of mathematics anxiety and academic performance: A basis for a proposed intervention program. International Journal of Multidisciplinary: Applied Business and Education Research. http://dx.doi.org/10.11594/ijmaber.06.11.03\

Vargas, R. (2021). A literature review on math anxiety and learning mathematics: A general overview. Journal of Educational Research and 102-108. Reviews, 9(5), https://doi.org/10.33495/jerr_v9i5.21.112

De La Hera, J. M. H., Morales-Rodríguez, F. M., Rodríguez-Gobiet, J. P., & Martínez-Ramón, J. P. (2023). Attitudes toward mathematics/statistics, anxiety, self-efficacy and academic

performance: an artificial neural network. Frontiers in Psychology, 14, 1214892. https://doi.org/10.3389/fpsyg.2023.1214892

Pizzie, R. G., & Kraemer, D. J. M. (2023). Strategies for remediating the impact of math anxiety on high school math performance. Npj Science of Learning, 8(1), 44. https://doi.org/10.1038/s41539-023-00188-5

c) Pisikal at emosyonal na salik (physical and emotional factors)

Wang, C., Li, X.-y., & Lin, X.-y. (2025). Peer support as a buffer: reducing math anxiety through psychological resilience in left-behind rural students. https://doi.org/10.1057/s41599-025-05075-5

Lunardon, M., Cerni, T., Zanon, M., & Rumiati, R. I. (2025). Physiological correlates of math ability in higher education. https://www.sciencedirect.com/science/article/abs/pii/S2211949325000201

Viray-Ariola, C. A. (2025). College Students' Level of Mathematics Anxiety and Academic Performance:A Basis for a Proposed Intervention Program http://dx.doi.org/10.11594/ijmaber.06.11.03

Daches Cohen, L., Korem, N., & Rubinsten, O. (2021). Math anxiety is related to math difficulties and composed of emotion regulation and anxiety predisposition: A network analysis study. Brain Sciences, 11(12), 1609.

Rada, E., & Lucietto, A. M. (2022). Math anxiety: A literature review on confounding factors. Journal of Research in Science, Mathematics and Technology Education, 5(2), 117–129.

Yan, X. (2024). How does math anxiety affect students' academic achievement in mathematics? —A Meta-Analysis based on 31 experimental studies. Journal of Social Systems and Policy Analysis, 1(4), 131–142. https://doi.org/10.62762/jsspa.2024.672218

Academic Performance

Siaw, E.S., Shim, G. T.G., Azizan, F. L., and Shaipullah, N. M. (2021). Understanding the relationship between students' math-ematics anxiety levels and mathematics performances at the foundation Level. Journal of Education and Learning, 10, (1), 21. https://doi.org/10.5539/jel.v10n1p47

Yakubu, J. Z., Bisandu, D. B., & Datiri, D. D. (2019). Relationship between Mathematics Test Anxiety and Achievement of Senior Secondary Three Students in Kafanchan Educational Zone, Kaduna State. June.

Trujillo-Torres, J. M., Hossein-Mohand, H., Gómez-García, M., Hossein-Mohand, H., & Hinojo-Lucena, F. J. (2020). Estimating the academic performance of secondary education mathematics students: A gain lift predictive model. Mathematics, 8(12), 1-23. https://doi.org/10.3390/math8122101

Samante, J. L., & Alave, A. D. (2021). Anxiety and Performance in Mathematics of Grade Four. 11(1), 1-6. https://doi.org/10.46360/globus.edu.220211001

Živković, M., Pellizzoni, S., Mammarella, I. C., & Passolunghi, M. C. (2022). Executive functions, math anxiety and math performance in middle school students. British Journal of Developmental Psychology, 40(3), 438-452 https://doi.org/10.1111/bjdp.12412

Namkung, J. M., Peng, P., & Lin, X. (2024). Math anxiety: Negative affect and cognitive beliefs linked to poor math performance. Contemporary Educational Psychology, 74, Article 102094, https://doi.org/10.1016/j.cedpsych.2023.102094

V. Pagpaplano ng Metodolohiya

a. Disenyo ng Pananaliksik – descriptive-correlational (quantitative research)
b. Respondente/Corpus – BSED Math 2nd year college students
c. Paraan ng Pagkalap ng Datos – Questionnaires (Likert Scale) and Grades sa Mathematics.
d. Paraan ng Pagsusuri – correlation

VI. Timeline/Work Plan

Ang timeline na ito ay saklaw lamang ang pagsagawa ng proposal hindi pa kasali ang final defense at paggawa ng pinal na Pampanahonang Papel.

DATE	GINAWA/GAGAWIN	TARGET OUTPUTS
01/24/26 – 01/25/26	• Pag-uusap sa Paksang Popokusan • Pagbuo ng Pamagat ng Pag-aaral • Pagbuo ng Layunin at mga Tanong • Pag-uusap sa Desinyo ng Pananaliksik • Pagiisip ng Posibleng Respondente ng Pag-aaral	• Paksa (Pinal) • Pamagat ng Pag-aaral (di-pinal) • Makabuo ng Layunin at mga Tanong • Magkaroon ng fixed na Disenyo ng Pananaliksik • College Students (BSEd Math)
01/26/26 – 02/01/26	• Paggawa ng Outline ng Proposal • Mangalap ng Sanggunian na Gagamitin sa Pag-aaral	• Proposal Outline • Related Literatures at Sanggunian nito
02/02/26 – 02/08/26	• Pagbuo ng Rasyunal ng Pag-aaral • Pagsagawa ng Contextualization ng mga Related Literatures/Studies	• Rasyunal/Introduksyon (draft) • Review of Related Literatures na seksyon sa Papel
02/09/26 – 02/15/26	Kalibulong Week	
02/16/26 – 02/22/26	• Pagbuo ng Pinal na Rasyunal ng Pag-aaral • Pagsagawa ng Pinal na Contextualization ng mga Related Literatures/Studies	• Rasyunal/Introduksyon (pinal) • Review of Related Literatures na seksyon sa Papel (Pinal)
02/23/26 – 03/01/26	Refine/Revise/Review ng Buong Proposal	Mas maayos na Proposal
03/02/26 – 03/08/26	Paghahanap ng Questionnaire	QUESTIONNAIRE (Likert Scale)
03/09/26 – 03/15/26	Magsagawa ng survey sa populasyon ng BSEd Math 2nd Year sa DNSC	Makakalap ng tamang populasyon ng 2nd year students
03/16/26 – 03/22/26	• Paghahanda sa Proposal Defense • Panahon na ng Proposal Defense	• Handa at Maayos na Pamanahonang Papel (Proposal) • Aprubado
03/23/26 – 03/29/26	Midterm Exam Week	

PANGKAT-1

BOCOBO, JUAN MIGUEL
CALUZA, ANGEL MARIE
GALENDEZ, HANNAH
IBARRA, ERIC JAMES
JUANITE, JHON MARQUIN
LIBRE, DAIROSE
LOON, BETHOVEN FRANZ
LOREMAS, QUEENALIZA
MABANDING, MARK KENNETH
MONTEALTO, RHYZA MAE
PAÑA, LUXER
SALINGAY, MIKE
SALPOCIAL, BRIAN